COMPOSITION OF INTERNAL HOM COMPLEXES AS A MAP FROM THE TOTAL TENSOR PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Internal Hom complex of two complexes and the total tensor product, with their sign conventions. discussed in the same section, source lines 17880-17964, before the counted statement. Supplies the complexes appearing in the statement and the sign rules used in the verification.
- Monoidal structure on complexes and the categorical composition map between internal Homs. the author cites Categories, Remark categories-remark-internal-hom-monoidal at proof lines 17981-17984. Explains the categorical source of the map.
- Finiteness of the sums involved in the components of an element of a total tensor product. used at proof lines 18015-18025 to show the image is well defined. Makes the componentwise assignment legitimate.

Source reference. Original lemma 0A8I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12916–12916. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0GWQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21080–21080. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0GWJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21074–21074. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Given complexes $K^\bullet, L^\bullet, M^\bullet$ of R-modules there is a canonical morphism*

$$Tot(\mathrm{Hom}^\bullet(L^\bullet, M^\bullet) \otimes_R \mathrm{Hom}^\bullet(K^\bullet, L^\bullet)) \longrightarrow \mathrm{Hom}^\bullet(K^\bullet, M^\bullet)$$

of complexes of R-modules.

Proof. Via the discussion in Remark 0GWQ the existence of such a canonical map follows from Categories, Remark 0GWJ. We also give a direct construction.

An element α of degree n of the left hand side is

$$\alpha = (\alpha^{p,q}) \in \bigoplus\nolimits_{p+q=n} \mathrm{Hom}^p(L^\bullet, M^\bullet) \otimes_R \mathrm{Hom}^q(K^\bullet, L^\bullet)$$

The element $\alpha^{p,q}$ is a finite sum $\alpha^{p,q} = \sum \beta_i^p \otimes \gamma_i^q$ with

$$\beta_i^p = (\beta_i^{r,s}) \in \prod\nolimits_{r+s=p} \mathrm{Hom}_R(L^{-s}, M^r)$$

and

$$\gamma_i^q = (\gamma_i^{u,v}) \in \prod\nolimits_{u+v=q} \mathrm{Hom}_R(K^{-v}, L^u)$$

The map is given by sending α to $\delta = (\delta^{r,v})$ with

$$\delta^{r,v} = \sum\nolimits_{i,s} \beta_i^{r,s} \circ \gamma_i^{-s,v} \in \mathrm{Hom}_R(K^{-v}, M^r)$$

For given $r + v = n$ this sum is finite as there are only finitely many nonzero $\alpha^{p,q}$, hence only finitely many nonzero β_i^p and γ_i^q. By our sign rules we have

$$\begin{aligned}
\mathrm{d}(\alpha^{p,q}) &= \mathrm{d}_{\mathrm{Hom}^\bullet(L^\bullet, M^\bullet)}(\alpha^{p,q}) + (-1)^p \mathrm{d}_{\mathrm{Hom}^\bullet(K^\bullet, L^\bullet)}(\alpha^{p,q}) \\
&= \sum \left(\mathrm{d}_M \circ \beta_i^p \circ \gamma_i^q - (-1)^p \beta_i^p \circ \mathrm{d}_L \circ \gamma_i^q \right) \\
&\quad + (-1)^p \sum \left(\beta_i^p \circ \mathrm{d}_L \circ \gamma_i^q - (-1)^q \beta_i^p \circ \gamma_i^q \circ \mathrm{d}_K \right) \\
&= \sum \left(\mathrm{d}_M \circ \beta_i^p \circ \gamma_i^q - (-1)^n \beta_i^p \circ \gamma_i^q \circ \mathrm{d}_K \right)
\end{aligned}$$

It follows that the rules $\alpha \mapsto \delta$ is compatible with differentials and the lemma is proved. □